

CHEM 121L: PRINCIPLES OF CHEMISTRY I LABORATORY



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CHEM 121L: Principles of Chemistry I Laboratory

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Detailed Licensing

Licensing

A detailed breakdown of this resource's licensing can be found in [Back Matter/Detailed Licensing](#).

CHAPTER OVERVIEW

1: Laboratory Equipment

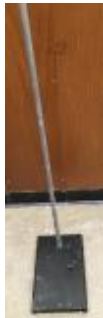
[1.1: Common Lab Equipment](#)

[1.2: Personal Lab Equipment](#)

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1.1: Common Lab Equipment

Common lab equipment can be found around the lab rather than in individual student drawers.

Ring Stand 	Iron ring 	Buret Clamp 
Labquest 	Colorimeter 	Cuvette 
pH electrode 	Conductivity electrode 	Temperature Probe 
Balance 	Hot plate/stir plate 	Magnetic Stir Bar 

Liquid waste container



Solid waste container



Ice Bath



Spectrophotometer



UV/Vis Cell Holder



Well plate



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1.2: Personal Lab Equipment

Laboratory Apron



Laboratory Glasses



Beakers



Erlenmeyer Flasks



Graduated Cylinders



Buret



Volumetric Pipets



Volumetric Flasks



Test Tubes and Test Tube Rack



Test Tube Holder



Watch Glass



Double Ended Spatula, Wooden Handled Spatula and Scoopula



Forceps



Stir Rod with Rubber Policeman



Filter Flasks



Filter Adapter



Long Stemmed, Buchner, and Buret Funnel



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CHAPTER OVERVIEW

2: Density and Graphing

Learning Objectives

- Use a graduated cylinder and buret properly.
- Read values on measuring instruments to the correct number of significant figures.
- Calculate with measured numbers and use the correct number of significant figures in your answer.

Introduction

Measurement

Measurements provide the macroscopic information that is the basis of most of the hypotheses, theories, and laws that describe the behavior of matter and energy in both the macroscopic and microscopic domains of chemistry. Every measurement provides three kinds of information: the size or magnitude of the measurement (a number); a standard of comparison for the measurement (a unit); and an indication of the uncertainty of the measurement. While the number and unit are explicitly represented when a quantity is written, the uncertainty is an aspect of the measurement result that is more implicitly represented and will be discussed later.

The number in the measurement can be represented in different ways, including decimal form and scientific notation. For example, the maximum takeoff weight of a Boeing 777-200ER airliner is 298,000 kilograms, which can also be written as 2.98×10^5 kg. The mass of the average mosquito is about 0.0000025 kilograms, which can be written as 2.5×10^{-6} kg.

Units, such as liters, pounds, and centimeters, are standards of comparison for measurements. When we buy a 2-liter bottle of a soft drink, we expect that the volume of the drink was measured, so it is two times larger than the volume that everyone agrees to be 1 liter. The meat used to prepare a 0.25-pound hamburger is measured so it weighs one-fourth as much as 1 pound. Without units, a number can be meaningless, confusing, or possibly life threatening. Suppose a doctor prescribes phenobarbital to control a patient's seizures and states a dosage of "100" without specifying units. Not only will this be confusing to the medical professional giving the dose, but the consequences can be dire: 100 mg given three times per day can be effective as an anticonvulsant, but a single dose of 100 g is more than 10 times the lethal amount.

We usually report the results of scientific measurements in SI units, an updated version of the metric system, using the units listed in Table 01.1. Other units can be derived from these base units. The standards for these units are fixed by international agreement, and they are called the International System of Units or SI Units (from the French, *Le Système International d'Unités*). SI units have been used by the United States National Institute of Standards and Technology (NIST) since 1964.

Table 01.1: Base Units of the SI System

Property Measured	Name of Unit	Symbol of Unit
length	meter	m
mass	kilogram	kg
time	second	s
temperature	kelvin	K
electric current	ampere	A
amount of substance	mole	mol
luminous intensity	candela	cd

Sometimes we use units that are fractions or multiples of a base unit. Ice cream is sold in quarts (a familiar, non-SI base unit), pints (0.5 quart), or gallons (4 quarts). We also use fractions or multiples of units in the SI system, but these fractions or multiples are always powers of 10. Fractional or multiple SI units are named using a prefix and the name of the base unit. For example, a length of 1000 meters is also called a kilometer because the prefix *kilo* means "one thousand," which in scientific notation is 10^3 (1 kilometer = 1000 m = 10^3 m). The prefixes used and the powers to which 10 are raised are listed in Table 01.2.

Table 01.2: Common Unit Prefixes

Prefix	Symbol	Factor	Example
femto	f	10^{-15}	1 femtosecond (fs) = 1×10^{-15} s (0.000000000000001 s)

Prefix	Symbol	Factor	Example
pico	p	10^{-12}	1 picometer (pm) = 1×10^{-12} m (0.000000000001 m)
nano	n	10^{-9}	4 nanograms (ng) = 4×10^{-9} g (0.000000004 g)
micro	μ	10^{-6}	1 microliter (μ L) = 1×10^{-6} L (0.000001 L)
milli	m	10^{-3}	2 millimoles (mmol) = 2×10^{-3} mol (0.002 mol)
centi	c	10^{-2}	7 centimeters (cm) = 7×10^{-2} m (0.07 m)
deci	d	10^{-1}	1 deciliter (dL) = 1×10^{-1} L (0.1 L)
kilo	k	10^3	1 kilometer (km) = 1×10^3 m (1000 m)
mega	M	10^6	3 megahertz (MHz) = 3×10^6 Hz (3,000,000 Hz)
giga	G	10^9	8 gigayears (Gyr) = 8×10^9 yr (8,000,000,000 Gyr)
tera	T	10^{12}	5 terawatts (TW) = 5×10^{12} W (5,000,000,000,000 W)

Density

We use the mass and volume of a substance to determine its density. Thus, the units of density are defined by the base units of mass and length.

The density of a substance is the ratio of the mass of a sample of the substance to its volume. The SI unit for density is the kilogram per cubic meter (kg/m^3). For many situations, however, this is an inconvenient unit, and we often use grams per cubic centimeter (g/cm^3) for the densities of solids and liquids, and grams per liter (g/L) for gases. Although there are exceptions, most liquids and solids have densities that range from about 0.7 g/cm^3 (the density of gasoline) to 19 g/cm^3 (the density of gold). The density of air is about 1.2 g/L . Table 1.6.31.6.3 shows the densities of some common substances.

Table 01.3: Densities of Common Substances

Solids	Liquids	Gases (at 25 °C and 1 atm)
ice (at 0 °C) 0.92 g/cm^3	water 1.0 g/cm^3	dry air 1.20 g/L
oak (wood) $0.60\text{--}0.90 \text{ g/cm}^3$	ethanol 0.79 g/cm^3	oxygen 1.31 g/L
iron 7.9 g/cm^3	acetone 0.79 g/cm^3	nitrogen 1.14 g/L
copper 9.0 g/cm^3	glycerin 1.26 g/cm^3	carbon dioxide 1.80 g/L
lead 11.3 g/cm^3	olive oil 0.92 g/cm^3	helium 0.16 g/L
silver 10.5 g/cm^3	gasoline $0.70\text{--}0.77 \text{ g/cm}^3$	neon 0.83 g/L
gold 19.3 g/cm^3	mercury 13.6 g/cm^3	radon 9.1 g/L

While there are many ways to determine the density of an object, perhaps the most straightforward method involves separately finding the mass and volume of the object, and then dividing the mass of the sample by its volume.

$$\text{Density} = \frac{\text{mass}}{\text{volume}} \quad (2.1)$$

Significant Figures in Measurement

Counting is the only type of measurement that is free from uncertainty, provided the number of objects being counted does not change while the counting process is underway. The result of such a counting measurement is an example of an exact number. If we count eggs in a carton, we know *exactly* how many eggs the carton contains. The numbers of defined quantities are also exact. By definition, 1 foot is exactly 12 inches, 1 inch is exactly 2.54 centimeters, and 1 gram is exactly 0.001 kilogram. Quantities derived from measurements other than counting, however, are uncertain to varying extents due to practical limitations of the measurement process used.

The numbers of measured quantities, unlike defined or directly counted quantities, are not exact. To measure the volume of liquid in a graduated cylinder, you should make a reading at the bottom of the meniscus, the lowest point on the curved surface of the liquid.

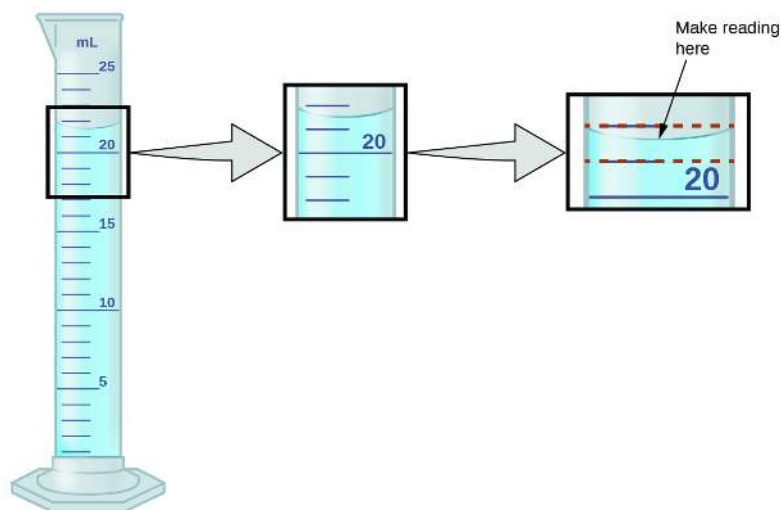
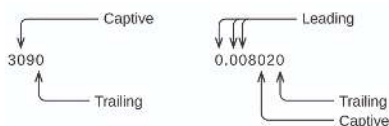


Figure 01.1: To measure the volume of liquid in this graduated cylinder, you must mentally subdivide the distance between the 21 and 22 mL marks into tenths of a milliliter, and then make a reading (estimate) at the bottom of the meniscus.

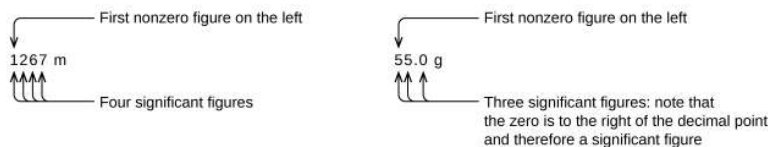
Refer to the illustration in Figure 01.1. The bottom of the meniscus in this case clearly lies between the 21 and 22 markings, meaning the liquid volume is *certainly* greater than 21 mL but less than 22 mL. The meniscus appears to be a bit closer to the 22-mL mark than to the 21-mL mark, and so a reasonable estimate of the liquid's volume would be 21.6 mL. In the number 21.6, then, the digits 2 and 1 are certain, but the 6 is an estimate. Some people might estimate the meniscus position to be equally distant from each of the markings and estimate the tenth-place digit as 5, while others may think it to be even closer to the 22-mL mark and estimate this digit to be 7. Note that it would be pointless to attempt to estimate a digit for the hundredths place, given that the tenths-place digit is uncertain. In general, numerical scales such as the one on this graduated cylinder will permit measurements to one-tenth of the smallest scale division. The scale in this case has 1-mL divisions, and so volumes may be measured to the nearest 0.1 mL.

This concept holds true for all measurements, even if you do not actively make an estimate. If you place a quarter on a standard electronic balance, you may obtain a reading of 6.72 g. The digits 6 and 7 are certain, and the 2 indicates that the mass of the quarter is likely between 6.71 and 6.73 grams. The quarter weighs *about* 6.72 grams, with a nominal uncertainty in the measurement of ± 0.01 gram. If we weigh the quarter on a more sensitive balance, we may find that its mass is 6.723 g. This means its mass lies between 6.722 and 6.724 grams, an uncertainty of 0.001 gram. Every measurement has some uncertainty, which depends on the device used (and the user's ability). All of the digits in a measurement, including the uncertain last digit, are called significant figures or significant digits. Note that zero may be a measured value; for example, if you stand on a scale that shows weight to the nearest pound and it shows "120," then the 1 (hundreds), 2 (tens) and 0 (ones) are all significant (measured) values.

Whenever you make a measurement properly, all the digits in the result are significant. But what if you were analyzing a reported value and trying to determine what is significant and what is not? Well, for starters, all nonzero digits are significant, and it is only zeros that require some thought. We will use the terms "leading," "trailing," and "captive" for the zeros and will consider how to deal with them.



Starting with the first nonzero digit on the left, count this digit and all remaining digits to the right. This is the number of significant figures in the measurement unless the last digit is a trailing zero lying to the left of the decimal point.

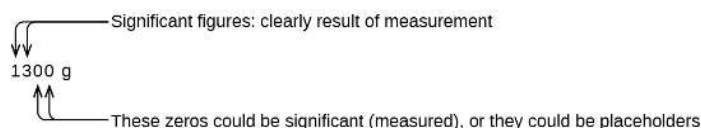


Captive zeros result from measurement and are therefore always significant. Leading zeros, however, are never significant—they merely tell us where the decimal point is located.



The leading zeros in this example are not significant. We could use exponential notation (as described in Appendix B) and express the number as 8.32407×10^{-3} ; then the number 8.32407 contains all of the significant figures, and 10^{-3} locates the decimal point.

The number of significant figures is uncertain in a number that ends with a zero to the left of the decimal point location. The zeros in the measurement 1,300 grams could be significant or they could simply indicate where the decimal point is located. The ambiguity can be resolved with the use of exponential notation: 1.3×10^3 (two significant figures), 1.30×10^3 (three significant figures, if the tens place was measured), or 1.300×10^3 (four significant figures, if the ones place was also measured). In cases where only the decimal-formatted number is available, it is prudent to assume that all trailing zeros are not significant.



When determining significant figures, be sure to pay attention to reported values and think about the measurement and significant figures in terms of what is reasonable or likely—that is, whether the value makes sense. For example, the official January 2014 census reported the resident population of the US as 317,297,725. Do you think the US population was correctly determined to the reported nine significant figures, that is, to the exact number of people? People are constantly being born, dying, or moving into or out of the country, and assumptions are made to account for the large number of people who are not actually counted. Because of these uncertainties, it might be more reasonable to expect that we know the population to within perhaps a million or so, in which case the population should be reported as 317 million, or 3.17×10^8 people.

Significant Figures in Calculations

A second important principle of uncertainty is that results calculated from a measurement are at least as uncertain as the measurement itself. We must take the uncertainty in our measurements into account to avoid misrepresenting the uncertainty in calculated results. One way to do this is to report the result of a calculation with the correct number of significant figures, which is determined by the following three rules for rounding numbers:

1. When we add or subtract numbers, we should round the result to the same number of decimal places as the number with the least number of decimal places (the least precise value in terms of addition and subtraction).
2. When we multiply or divide numbers, we should round the result to the same number of digits as the number with the least number of significant figures (the least precise value in terms of multiplication and division).
3. If the digit to be dropped (the one immediately to the right of the digit to be retained) is less than 5, we “round down” and leave the retained digit unchanged; if it is more than 5, we “round up” and increase the retained digit by 1; if the dropped digit is 5, we round up or down, whichever yields an even value for the retained digit. (The last part of this rule may strike you as a bit odd, but it’s based on reliable statistics and is aimed at avoiding any bias when dropping the digit “5,” since it is equally close to both possible values of the retained digit.)

The following examples illustrate the application of this rule in rounding a few different numbers to three significant figures:

- 0.028675 rounds “up” to 0.0287 (the dropped digit, 7, is greater than 5)
- 18.3384 rounds “down” to 18.3 (the dropped digit, 3, is less than 5)
- 6.8752 rounds “up” to 6.88 (the dropped digit is 5, and the retained digit is even)
- 92.85 rounds “down” to 92.8 (the dropped digit is 5, and the retained digit is even)

Pre Lab Video



Procedure

Buret:

Step 1



Clean and dry a 100 mL beaker. Weigh the beaker. **Note: You should use the same balance every time in this experiment.** Record the mass of your beaker.

Step 2



Fill the buret to approximately 0 mL with sodium chloride solution. You do not need to have it exactly at 0.00mL, but somewhere close and below the mark.

Step 3

Open the stop cock and allow it to fill with solution. Read the volume and record this as the initial volume (remember to read correct number of significant figures).



Step 4



Allow approximately 15 mL of your sodium chloride solution sample into the pre-weighed beaker. Record the volume as the final volume. Note: burets are read from top down, so the value should be somewhere around 15.

Step 5



Weigh the beaker and solution. Record the mass.

Step 6



Leaving the sodium chloride solution that you just measured in the beaker, dispense approximately 15 mL more solution into the beaker using your buret, being sure to record the initial and final volumes. Again, the final volume should have a value around 30. **DO NOT REFILL THE BURET.** Reweigh the beaker and sodium chloride solution. Record its mass in your notebook.

Step 7

Repeat step 3 so that you have three data points. Dispose of the sodium chloride solution sample down the drain.

Graduated Cylinder:

Step 1



Dry your beaker and reweigh it, being sure to record the mass in your notebook. Do not assume the mass is the same as before.

Step 2

Measure approximately 15 mL of your sodium chloride solution in a 100 mL graduated cylinder. Read the volume to the correct number of significant figures and record it in your notebook.



Step 3



Pour the sodium chloride solution into your beaker and weigh the beaker and solution together. Record the mass in your notebook.

Step 4



Leaving the sodium chloride solution you just measured in the beaker, repeat steps 2 and 3 twice more.

Calculations:

Step 1

Make a table of your data. You should have three measurements of sodium chloride solution and beaker using the buret and the graduated cylinder.

Step 2

Calculate (using the correct number of significant figures) the mass of sodium chloride solution added in each step.

Step 3

Calculate (using the correct number of significant figures) the density of sodium chloride solution using the equation

$$D = \frac{m}{V} \quad (2.2)$$

where m=mass in grams and V=volume in mL

Step 4

Do this for each measurement that you made and using your exchanged data. You should have 6 density measurements.

How does the density you measured compare to that of water?

Graphing Practice:

Part 1:

You are given the following results from a sample carbon dioxide gas under standard pressure (1atm).

Volume (L)	Temperature (°C)
24.7	35.49
32.1	91.72

34.8	149.95
40.6	218.51
46.2	281.51
50.3	331.88

You also want to know the volume at two other temperatures but were unable to obtain this data directly from the experiment. These temperatures are 220 °C and 335 °C.

Step 1

Make a graph in Excel using the volume and temperature data, including the equation and the R^2 value. See directions below. Make one graph choosing a trendline that is linear and print a copy. Make a second graph using the data but now choose a trendline that is an exponential fit. Finally, make a third graph with the data and choose a power trend line. Be sure to print copies of all three graphs. Based on the R^2 value which trendline gives the best fit?

Step 2

Using the best curve provided by Excel and the equation on the graph, what is the volume of product at 220 °C, at 335 °C?

Part 2:

The following are results from an experimental determination of the volume of nitrogen at various pressures at room temperature (298 K).

Volume (mL)	Pressure (atm)
41.6	0.526
32.7	0.658
26.2	0.789
24.3	0.921
21.6	1.053
18.9	1.184
17.4	1.316
15.6	1.447
13.9	1.579

Step 1

Graph this data using Excel. Try several trendlines and print the best version. How does this graph compare with the best graph for Part A? Explain your observations.

Step 2

What would your volume be at a pressure of 1.400 atm? 2.100 atm?

Part 3:

Step 1

Using your data, create a graph plotting the mass vs. volume for both the buret data and the graduated cylinder data. Be sure to use the total volumes and masses for each run. Please choose a linear trendline.

Step 2

Given the equation, what does the slope of the line represent?

Directions for Making a Graph in Excel



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CHAPTER OVERVIEW

3: Spectroscopy

Learning Objectives

- Use spectroscopy to examine atomic structure
- Gain an understanding about the relationships between energy and light.

Introduction:

A source of light (often called a continuous source) that has all of the different colors (wavelengths) and is called white light. However, some sources of light have only certain colors (discrete wavelengths) as components and appear colored to the eye.

Spectroscope: A spectroscope splits visible light into its component wavelengths (colors) and uses the human eye as the detector.

Spectrophotometer: A spectrophotometer splits light into its component wavelengths (colors) and uses an electronic detector.

This experiment explores the spectra of fluorescent light wavelengths using the spectroscopes and of various lamps containing gases with discrete wavelengths of radiation using the spectrophotometer.

Pre Lab Video



Experiment:

Part I: Spectroscopes and Tubes

Step 1



Obtain a spectroscope. Look into the spectroscope while aiming it at the overhead lights.

Step 2

You should see a numbered scale to the right with numbers ranging from four to seven. Underneath that you should see a series of broad swaths of colors from red to purple (see Figure 1). The scale is listed in hundreds of nanometers, so the “4” refers to 400 nm. This is the wavelength of that color of light.

Step 3

Draw the spectra in your lab notebook.

Step 4

Now turn on the hydrogen lamp and do the same. As soon as you are finished with the lamp though, turn it off. The gas lamps are very easy to burn out if left on too long.

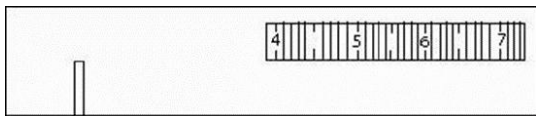


Figure 3.1: Inside of the Spectroscope

Part II: Using a spectrophotometer to record wavelengths

Each lamp has a spectrophotometer next to it connected to a computer. The spectrophotometer is an instrument that accurately measures the wavelengths and intensities of light.

Step 1

On the computer open SI 400 Series Emission.

Step 2

Click on the Initialize button.

Step 3



Holding the end of the fiber optic against a piece of clothing (to block light from entering) click on the “Dark” button on the screen.

Step 4



Hold the end of the fiber optic cable up to the lamp. Click on the “Acquire” button on the screen (best done by second person). Continue to hold the end of the fiber optic cable to the lamp until the spectra appears in the window.

Step 5

Examine the spectrum. You should see a set a plot (wavelength in nanometers on the x-axis and intensity on the y-axis) with a set of peaks. Each peak represents a different wavelength of light.

Step 6

Determine the wavelengths of the peaks. Use the arrow to grab the blue cursor and move it to the top of a peak. Click on the “Blue Cursor” button on the screen. The wavelength and intensity of that peak will appear in the box. Repeat until the major peaks have all been identified. Do this for up to the eight strongest peaks.

Step 7

Enter the appropriate information in the boxes for Sample name, Comment and Operator.

Step 8

Print your spectrum by clicking on the “Print” button on the screen the number of times as there are members in your group.

Step 9

Pick up your printouts.

Step 10

Complete both the hydrogen lamp and the other lamp at your initial station. Then move to the next station until you have the spectrum for each of the other remaining lamps.

Post Lab

From the data on your hydrogen spectrum from the spectrophotometer, calculate the energy associated with one of the three wavelengths of light using Planck's constant (h), the speed of light (c) and your wavelength λ .

$$E = \frac{hc}{\lambda} \quad (3.1)$$

Determine the color associated with light of this wavelength. Is this value consistent with any of the light you observed for the hydrogen lamp using the spectroscope?

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CHAPTER OVERVIEW

4: Halogens

Learning Objectives

- Determine how halogens (Cl_2 , Br_2 , I_2) react with halides (Cl^- , Br^- , I^-)?
- Explain the reactivity taking place.

Introduction:

Before you start the experiment, there are a couple of terms you should understand.

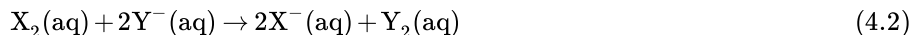
Halogen – The elements in Group 17 (or VIIA or 7A) of the periodic table are called the halogens. These elements are naturally found as diatomic species. In this experiment, they are (Cl_2 , Br_2 , and I_2). Cl_2 is a gas, Br_2 is a liquid and I_2 is a solid at room temperature. We have made solutions of all of these in water.

Halide ion– Halides are any of the negatively charged ions of the halogens. In this experiment, we are using the sodium salts NaCl , NaBr and NaI to provide the halide ions. When dissolved in water, these salts dissociate into the sodium ion and the halide ion. For example:



For this experiment, the sodium ion (Na^+) does not participate in the chemical reactions.

The chemical reactions of interest are symbolized by the equation



Pre Lab Video:



Safety and Waste Disposal

- The solutions must be disposed of in the hazardous waste container.

Experiment:

Warning: Make sure that you are using the correct solutions as you conduct this experiment. Keep your test tubes labeled throughout the experiment.

Part I: Solubility of the halogens and halides

In this part of the experiment we will be testing the solubility of the halides and halogens in both water and cyclohexane. This information will be used to draw conclusions about the results in part II.

Step 1



Label six small test tubes; two NaCl, two NaBr, and two NaI.

Step 2



Test the solubility of the halides in water and cyclohexane. Add a small amount (spatula tip) of solid NaCl, solid NaBr and solid NaI to the separate test tubes and then add ~1.0 mL of water. Mix thoroughly. Record your observations including color and clarity of the solutions including any solids present. Keep these solutions.

Step 3



Test the solubility of the halides in water in cyclohexane. Repeat step B in the other test tubes with ~1.0 mL of cyclohexane rather than water. Record your observations including color and clarity of the solutions and the presence of any solids. Keep these solutions.

Step 4



Combine the NaCl aqueous solution prepared in step B with the tube containing NaCl and cyclohexane prepared in C. Mix vigorously for 15 seconds. Record your observations. Repeat for the bromide and iodide solutions. Describe what happens to the solid and the solutions.

Step 5



Obtain about 4 mL each of the pre-made aqueous halogen solutions of Cl₂, Br₂, and I₂, and place in three labeled test tubes. **CAP UP** with parafilm. Keep the parafilm on when not using the solutions.

Step 6

Add 12-15 drops of each aqueous solution to separate labeled test tubes containing 1.0 mL of cyclohexane. Mix vigorously. Record your observations of both, the aqueous halides solutions (before adding the cyclohexane) and after mixing with cyclohexane. *Save the excess of all the solutions for Part II.*

Step 7

Summarize your results in a table.

Step 8

Before continuing describe briefly in your notebook how you will use the color of the cyclohexane layer to determine what halogen is present. Confer with your lab instructor or TA and have them initial your notebook.

Part II: Reactions between the halogens and halides

In this section we will be examining the reactions of each halogen with each different halide. That is, you should have a reaction of bromine with chloride and iodide, a reaction of chlorine with bromide and iodide, and a reaction of iodine with chloride and bromide.

Note

Be sure you are using the correct solutions and label every test tube. In this part, use the pre-made *sodium halide* solutions available.

Step 1



Add ~8 drops of aqueous Br_2 solution to 1 mL of aqueous chloride (Cl^-) solution. Mix thoroughly. Add 0.5 mL of cyclohexane and mix vigorously. Record your observations.

Does the cyclohexane indicate that a reaction between Cl^- and Br_2 occurred? Defend your conclusion based on the experimental data you obtained and the data from Part I.

Step 2

Add 8 drops of aqueous Br_2 solution to 1 mL of aqueous iodide (I^-) solution. Mix thoroughly. Add 0.5 mL of cyclohexane and mix vigorously. Record your observations.

Step 3

Add 8 drops of aqueous Cl_2 solution to 1 mL of aqueous bromide (Br^-) solution. Mix thoroughly. Add 0.5 mL of cyclohexane and mix vigorously. Record your observations.

Step 4

Add 8 drops of aqueous Cl_2 solution to 1 mL of aqueous iodide (I^-) solution. Mix thoroughly. Add 0.5 mL of cyclohexane and mix vigorously. Record your observations.

Step 5

Add 8 drops of aqueous I_2 solution to 1 mL of aqueous chloride (Cl^-) solution. Mix thoroughly. Add 0.5 mL of cyclohexane and mix vigorously. Record your observations.

Step 6

Add 8 drops of aqueous I_2 solution to 1 mL of aqueous bromide (Br^-) solution. Mix thoroughly. Add 0.5 mL of cyclohexane and mix vigorously. Record your observations.

Step 7

Discard all wastes in waste containers.

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CHAPTER OVERVIEW

5: Lewis Structures

Note: This PRE-LAB ASSIGNMENT MUST BE COMPLETED BEFORE COMING TO LAB. In your lab notebook, draw a large picture (Lewis structure) of all the molecules (such that you can write distances above the bond). You will use these structures label bond lengths and angles as you proceed through the experiment.

Learning Objectives

- To learn how to use one type of computer modeling
- To look at what affects bond lengths and bond angles.

Experiment:

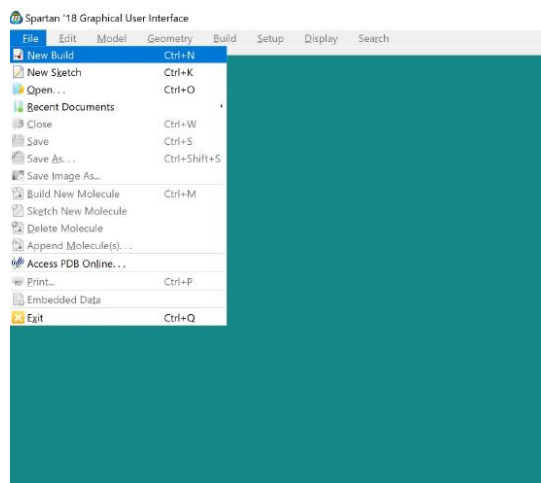
We will be using a program called Spartan to construct and examine molecules in during lab.

Part I:

Construct the molecules in the table below using Spartan. Minimize each structure after constructing it, by using the *Minimize* button.  This must be completed after building each molecule.

Bond Order	Terminal Atoms	Hetero Atoms
CH ₄	CF ₄	H ₃ CNH ₂
H ₃ C – CH ₃	CCl ₄	H ₃ COH
H ₂ C = CH ₂	CBr ₄	H ₃ CPH ₂
HC ≡ CH	CI ₄	H ₃ CSH

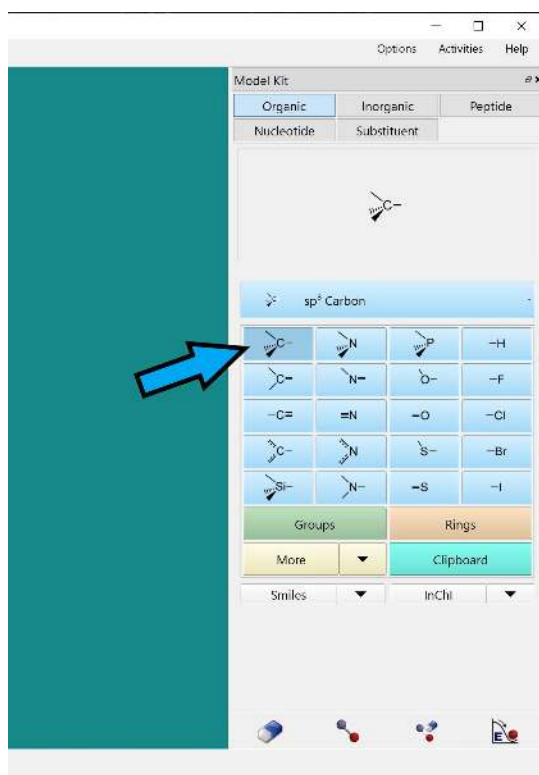
Step 1



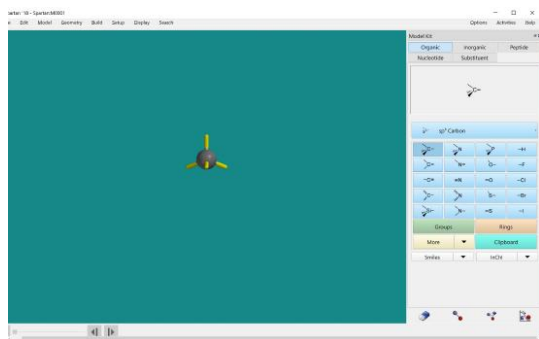
Open Spartan and click on new build.

Step 2

Click on the carbon atom with four bonds.

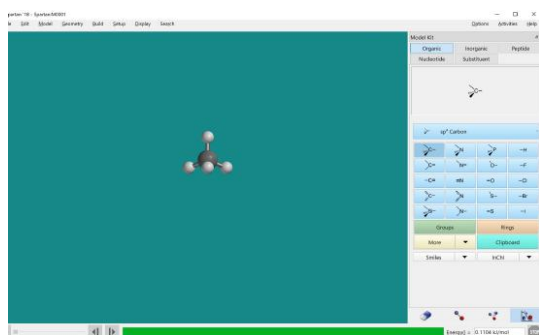


Step 3



Double click on the screen. You will see a carbon atom with four bonds appear.

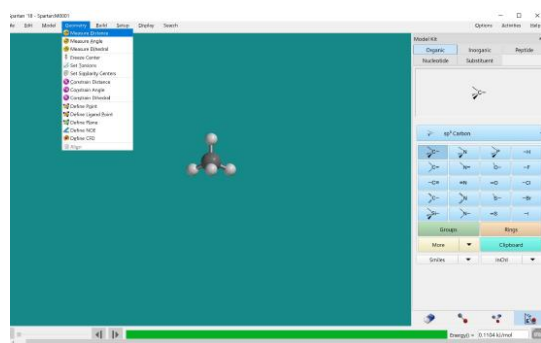
Step 4



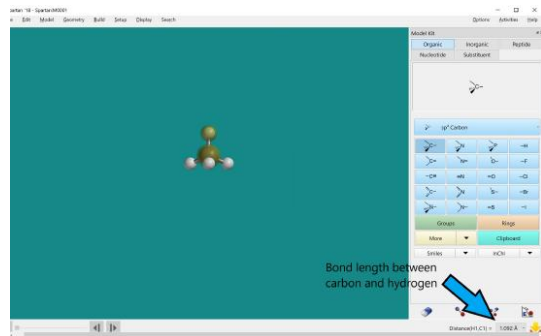
Click the minimize button. This will place hydrogen atoms on any open bonds. Reminder this must be completed after each and every molecule is built regardless if hydrogen needs to be added or not.

Step 5

Click measure distance under the Geometry menu.

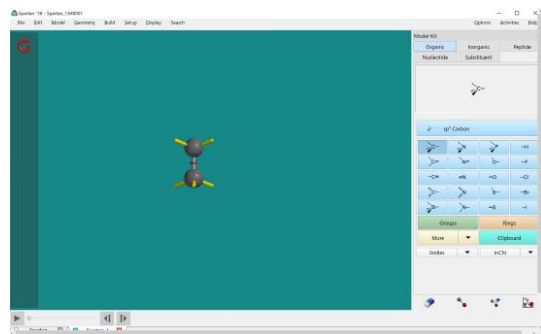


Step 6



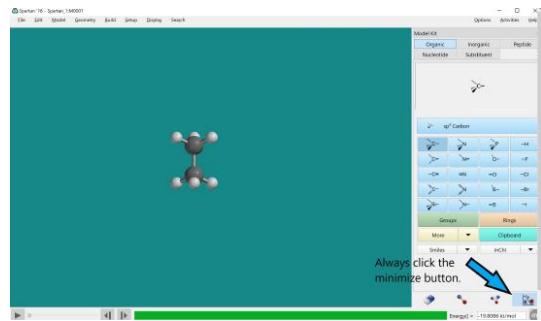
Click the two atoms on either side of the bond that you would like to measure or click on the bond. The distance measurement will be in the lower right hand corner.

Step 7



To create a bond between two carbons, double click the end of one of the bonds.

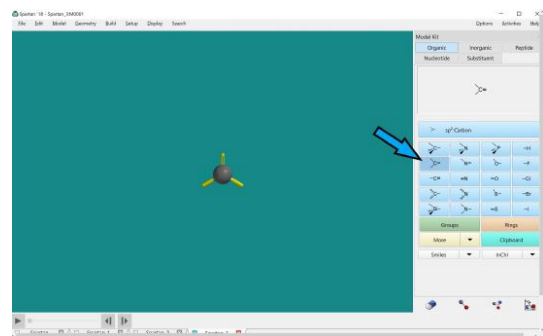
Step 8



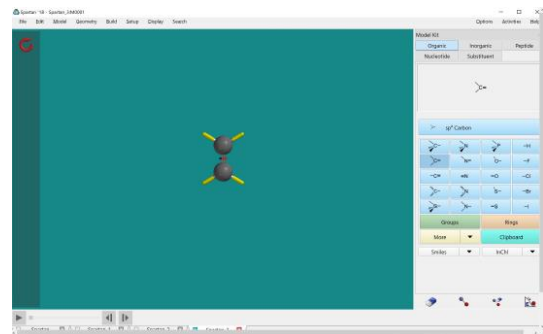
Click the minimize button. Measure bond lengths.

Step 9

To create a double bond between two carbon atoms, click the carbon with the double bond in the atom menu. Double click on the screen.

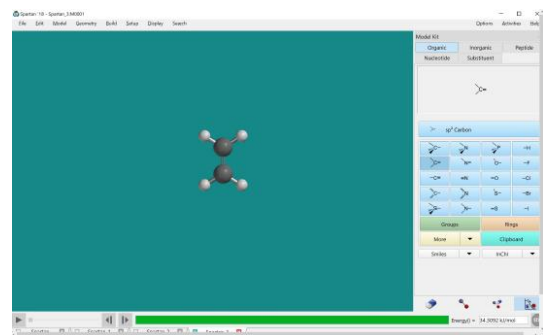


Step 10



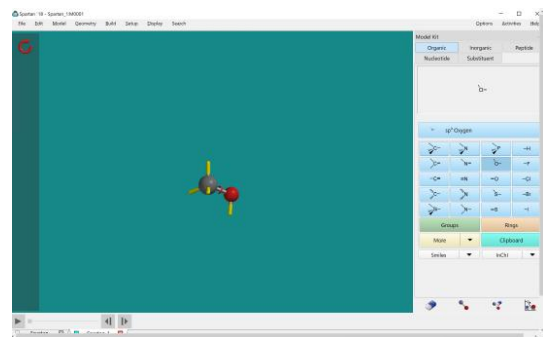
To attach the second carbon atom, double click the end of the double bond.

Step 11



Click minimize. Measure bond lengths.

Step 12



To attach a different atom to a carbon, first click the carbon with four bonds and then double click on the screen. Then click the atom you would like to attach in the atom menu and double click the end of one of the bonds on the carbon that you just created on the screen.

Application of Part I:

Step 1

Using the methods discussed in the pre-lab and your data from Part I-1, predict the lengths of the following bonds. Explain your answers.

N-F, N-O, N-Cl, H-F

Step 2

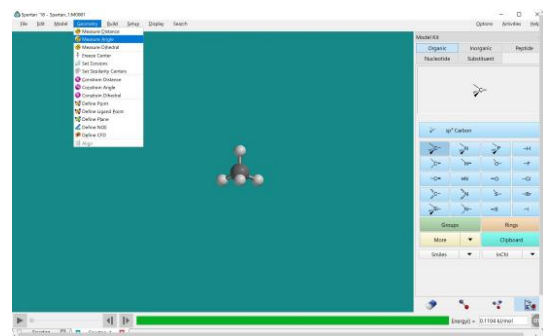
Construct a molecule containing one or more of these bonds so that you can measure the bond lengths on Spartan. You must be certain that you can compare the bonds. This means that they must all have the same bond order (all single bonds). Do your predictions agree with Spartan?

Part II:

Construct the following molecules and measure all the bond angles in the molecule. Be sure that you select the atoms in the order of the angle being measured. i.e. H – C – H for CH₄. Record these drawings and your bond angles in your notebook.

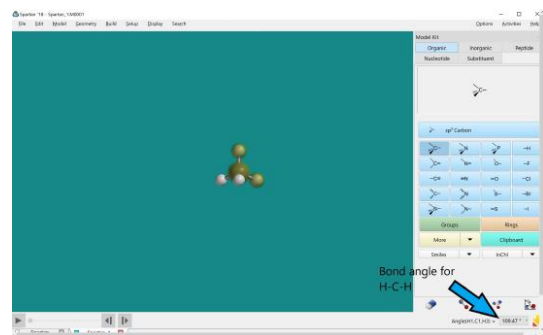
CH ₄	NH ₃	H ₂ O
H ₂ C = CH ₂	H ₂ C = O	HC ≡ CH
		

Step 1



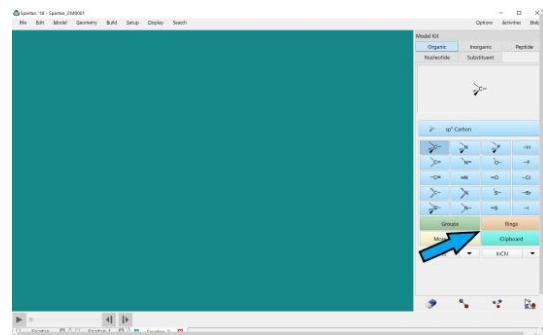
Build CH₄ molecule. To measure the angle, click measure angle under the geometry menu.

Step 2



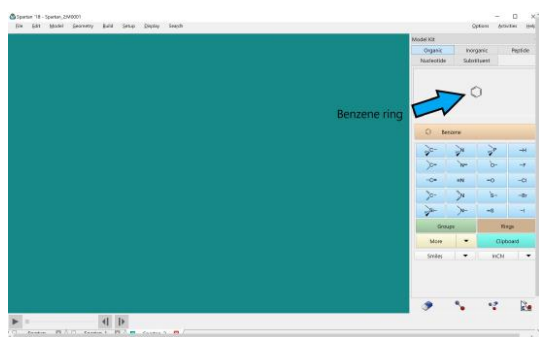
Click the atoms in the proper order, hydrogen then carbon followed by the last hydrogen in the bond angle to be measured. The bond angle will be in the lower right hand corner.

Step 3



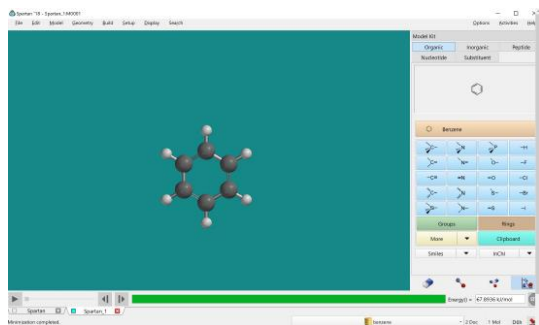
To build a benzene ring, click rings in the atom menu.

Step 4



The benzene ring will appear in the top box.

Step 5



Double click on the screen. Click the minimize button and measure the angles of the carbon, carbon, carbon bond as well as the hydrogen, carbon, carbon bond.

Part III:

Some practice on Lewis Structures

Complete the steps below for the molecules, NO_2^+ N_2O NO_2Cl NH_2^- Note: the underlined atom is the central atom.

Step 1

Draw the complete, correct Lewis structure

Step 2

Write all formal charges that are different from zero (0)

Step 3

Indicate the total number of electronic domains around the central or all internal atom(s) (underlined), specifying whether they are bonding or non-bonding.

Step 4

Predict the electronic and molecular shapes of the central atom in the ions and molecules below. It is up to you to determine if you use multiple bonding of the oxygen atoms to the nitrogen atom but you must explain your choices in your report.

Discussion:

- Summarize your data for the experiment in a table.
- What factors control the length of a bond?
- What factors control the angle of a bond?
- Did you see any relationship between bond order and bond angle?

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CHAPTER OVERVIEW

6: Empirical Formula

Learning Objectives

- To determine the empirical formula of a compound.

An empirical formula of a compound is the simplest whole number ratio of the various atoms in a compound. In this experiment, you will determine the empirical formula of the compound that results when magnesium and oxygen react.

Pre Lab Video



Using a Bunsen Burner

Bunsen burners are a source of heating in the laboratory. However, care must be taken when using a Bunsen burner.

Prior to using:

1. Know where all safety equipment is, especially the fire extinguisher.
2. Inspect burner for any damage and tubing for any cracks and connect the tubing to gas valve making sure there is a tight fitting connection.
3. Remove any papers, notebooks, etc. from immediate working area.
4. Tie back and long hair.
5. Make sure to be able to get a spark using a striker.

When lighting the burner:

1. Turn on gas.
2. Use striker to light burner.
3. If burner does not ignite within 30 seconds, turn off gas outlet and practice using the striker again.
4. Once burner is lit, adjust the gas valve to control the size of the flame.
5. Adjust the air inlet to achieve a blue flame with an inner blue cone.

During use of burner:

1. Never leave the burner unattended.
2. If the flame goes out, turn off gas and wait 30 seconds before reigniting.
3. Make sure heated glassware is open and pointed away from any individuals including yourself.

After use:

1. Turn off gas as soon as you are finished using it.
2. Heated materials will be hot even if they do not look like it. Use extreme caution before touching anything that has been heated.
3. Handle heated materials with crucible tongs or heat resistant gloves. This includes ring stands, rings, clay triangles, etc.
4. Set hot glassware on hot pads or trivets.
5. Allow materials to cool before cleaning up.

Using a Crucible

Crucibles are used to heat solids in the lab. However, handle crucibles with care as they are fragile.

Before using a crucible:

1. Check surface for cracks or imperfections.
2. Set up ring and clay triangle on a ring stand.
3. Place crucible on clay triangle ensuring a good fit.

During use:

1. Light burner away from ring and clay triangle.
2. Move burner under crucible to heat.
3. Heat gently or strongly based on instructions. To heat gently, maintain distance between the flame and the crucible. To heat strongly, adjust burner such that the inner blue cone on the flame is touching the bottom of the crucible.
4. Adjust crucible lid for appropriate oxygen levels. Cover tightly to keep oxygen away from sample. Open crucible lid or tilt lid on crucible to allow oxygen to come in contact with sample.
5. Use crucible tongs move crucible and lid.

After use:

1. Use care when moving the crucible. It will be hot, and it is fragile.
2. Place crucible on a hot pad or trivet.
3. Do not mass crucible until it is completely cool.
4. Be sure gas has been turned off on the Bunsen burner.

Pre-Lab Assignment

In your notebook, explain why sometimes a crucible cover is placed such that it fits tightly on the crucible during heating and other times it is slightly tilted.

Safety and Waste Disposal

- The solids must be disposed of in the hazardous waste container.

Procedure

Safety glasses must be worn at all times.

If the magnesium flashes during heating, do not look directly at the bright light emitted (it could damage your eyes)

Step 1



Clean a crucible and lid, rinsing thoroughly with deionized water as a last step.

Step 2



Place the clean and dry crucible and cover on a clay triangle and heat strongly (flame directly under the crucible and touching the bottom) for 2 minutes to drive off any volatile material.

Step 3

While the crucible is heating, clean thoroughly Mg ribbon weighing approximately 0.3 grams with sandpaper to remove any oxide coating. Avoid handling the ribbon with your finger; this will leave deposits on the Mg ribbon.



Step 4



Using crucible tongs remove the crucible and cover from the clay triangle and place them on a square pad from your drawer to cool.

Step 5



Allow the crucible to cool and weigh the crucible and lid. Handle the crucible with tongs, so you do not leave any deposits from your fingers. Record the weight in your lab notebook.

Step 6

While the first crucible is cooling complete Steps 1 – 3 on a second crucible.

Step 7



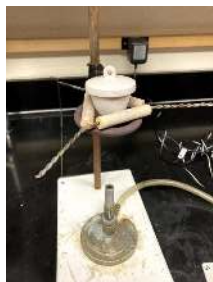
Coil the ribbon, without touching it, very loosely and place on the bottom of the crucible. Then, weigh the crucible and lid with the Mg ribbon inside. Record the second weight in your lab notebook.

Step 8



Heat the crucible gently (burner off to the side but nearly touching the crucible) for 2 minutes while using the tongs to lift the cover slightly every 10 sec. to admit air. Should the Mg start glowing brightly when the cover is lifted, cover the crucible, for 10 – 15 seconds. Resume lifting the lid.

Step 9



Heat the covered crucible strongly for 5 minutes continuing to lift the cover occasionally.

Step 10

Lift the cover to determine whether the ribbon has become a whitish ash. If the ribbon is not white ash after 5 minutes, inform your instructor. Heat for a total of 10 minutes ensure the ribbon is no longer reigniting. Allow the crucible to cool. *While it is cooling, you should calculate the moles of magnesium you added to the crucible.*

Step 11

While the first experiment crucible is cooling complete Steps 7 – 10 with the second crucible and magnesium sample.

Step 12



After the crucible is at room temperature, add 10 drops of distilled water.

Step 13



Partially cover the crucible (leave a slight crack) and heat gently for 1 minute, then strongly for 1 minute. Allow the crucible to cool to **room temperature**. While the first crucible is cooling complete Steps 12 & 13 with the second crucible.



Step 14



Weigh the crucible with product.

Step 15

Reheat strongly (2 minutes), cool and weigh. Repeat this process if the difference between the mass and the mass in number 14 has a difference greater than 0.05 g. *Do not dispose of your sample until your instructor approves your masses.

Step 16

Dispose the product in the waste container labeled "waste magnesium oxide".

Calculations

From the data you have collected, you should be able to find the amount of magnesium in the sample. You should also be able to find the mass of the oxygen that has reacted with the magnesium. From these two masses and the molar masses of magnesium and oxygen, you can calculate the number of moles of magnesium and oxygen present in the final product. Then, you can calculate the empirical formula of the magnesium oxide you made.

Step 1

Calculate the mass of the magnesium added to the crucible.

Step 2

Calculate the mass of oxygen added by finding the difference between the crucible with the final product and the crucible with the magnesium.

Step 3

Calculate the moles of magnesium and oxygen in the product.

Step 4

Determine the empirical formula for magnesium oxide.

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CHAPTER OVERVIEW

7: Limiting Reactant

Learning Objectives

- To use observations to determine the correct reaction and stoichiometry.
- Determine the limiting reactant for different mixtures of the reactants.
- Use mole-mass relationships to calculate which reactant will be limiting.

Pre Lab Video



Separations

Solids are sometimes produced when a chemical reaction takes place. We call these solids a precipitate. There are several ways to separate a solid from a solution. Below you will find two different ways we will be using in lab.

Decanting

When decanting, the precipitate is given time to settle to the bottom of the container. Once the solid has settled, the liquid, or supernatant, is carefully poured out of a beaker leaving the solid behind.

Filtration

When a precipitate is present in a solution, we will use one of two types of filtration in our lab, gravity and suction filtration. During gravity filtration, filter paper is folded to fit into a long stem funnel by folding the filter paper in half and then in fourths (Figure 1 and 2). The filter paper is then opened such that one half has three thickness of filter paper and the other half has one (Figure 3). The filter paper is placed in the funnel and water is added to the filter paper to keep it in place (Figure 4). Note there should be a container placed under the funnel stem to catch the water and filtered solution, or filtrate. Decant as much of the solution as possible before pouring the solid in the funnel. This will increase the speed of filtration. Once the solid is transferred to the funnel, use a wash bottle to rinse the remaining solid from the original container to the funnel. Rinse several time to collect as much solid as possible (Figure 5).



Figure 7.1: Folding the filter paper in half



Figure
7.2

: Folding the filter paper in fourths

Figure 7.3: Opening the filter paper



Figure
7.4

: Wetting the filter paper

Figure 7.5: Final rinse of beaker.

Safety and Waste Disposal

- The solutions must be disposed of in the hazardous waste container. Filter paper must be disposed of in the solid waste container.

Experiment:

This experiment has more experimental detail than previous experiments. However, you need to pay attention to what occurs in each step along the way. There are some questions embedded in the procedure that you need to answer before continuing on to the next step. These should be answered in your lab notebook, as they will help you with your discussion later. The questions are indicated in bold type. Be sure to answer all of them as you go.

Part I: What Happens when we mix cobalt (II) nitrate and sodium phosphate solutions together?

Answer all questions in your laboratory notebook as you complete this experiment.

Step 1



Dissolve a small amount of cobalt (II) nitrate, $\text{Co}(\text{NO}_3)_2$, in about 20 mL of distilled water. Dissolve a similar amount of sodium phosphate, Na_3PO_4 , in a second 20 mL of water. Describe the appearance of each solution.



Step 2

What ions are dissolved in each solution?

Step 3

Pour half of each solution into a third beaker and mix thoroughly. (Save the mixture and two solutions for later use.) Describe the appearance of the mixture.

Step 4

Assuming the reaction involves the coming together of dissolved ions, what are the possible identities of the solid formed in the mixture?

Step 5



Separate the mixture by gravity filtration. Describe the characteristics of the liquid, called the *filtrate*. What are the possible identities of the ions in the solution?

Step 6

Divide the filtrate in half and test each half with the remaining $\text{Co}(\text{NO}_3)_2$ and Na_3PO_4 solutions. Describe the results.

Step 7

Based on your results in part 6, identify the ions in the solution and the identity of the solid formed. Write a balanced chemical equation based on these results.

Step 8

Discuss your ideas with your lab instructor before continuing.

Part II: Can we tell which reactant runs out first?

Step 1

Record the concentrations of the $\text{Co}(\text{NO}_3)_2$ and Na_3PO_4 solutions in your notebook.

Step 2

Clean and rinse with distilled water four 100 mL beakers and label them 1-4.

Step 3



Into each beaker add 20.0 mL of stock $\text{Co}(\text{NO}_3)_2$ solution. Add different amounts of stock Na_3PO_4 into each beaker. The suggested amounts are: 5.0, 10.0, 15.0, and 20.0 mL. Using the appropriate graduated cylinder, record exact the volumes of each solution you added in your table. Mix each solution thoroughly and allow it to stand for at least 5 minutes.



Step 4

While you are waiting, calculate the mass of $\text{Co}(\text{NO}_3)_2$ added to each beaker and record these amounts in the table in your notebook. Calculate the mass of Na_3PO_4 added to each beaker and record this in your notebook.

Step 5

Set up four filter funnels with filter papers and labeled collection beakers. Decant and filter each reaction mixture from part II through a separate funnel, collecting its filtrate in a labeled beaker. You are to collect 10-20 mL of each filtrate, clean of any

precipitate. If the filtrate is cloudy, refilter your solution.

Step 6

Describe the appearance of each of your filtrates.

Step 7

Divide each of the filtrate into two parts. Test one part with a dropperful of stock $\text{Co}(\text{NO}_3)_2$ solution and the other with a dropper full of stock Na_3PO_4 solution. Summarize the results of these tests in your notebook.

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CHAPTER OVERVIEW

8: Hess's Law

Learning Objectives

- To determine the enthalpy, ΔH , of several reactions experimentally.
- To use Hess's Law to determine the enthalpy, ΔH , for a reaction

The study of energy is a vital function of chemistry. Over 90 percent of the energy we produce comes from chemical reactions such as the combustion of fossil fuels. This study of energy and its transformations is known as thermochemistry. As chemical bonds break and form in a chemical reaction, energy in the form of heat is either released or absorbed. Under conditions of constant pressure, the energy change of a reaction is called the heat of reaction or enthalpy change, ΔH . The amount of heat either absorbed or released during a reaction can be measured if the reaction is carried out in a container that insulates the reaction from its surroundings. By convention if heat is released, the reaction is termed exothermic and ΔH is negative. If heat is absorbed, the reaction is termed endothermic and ΔH is positive. In this lab, you will determine experimentally the value for ΔH for the acid-base neutralization reaction involving hydrochloric acid and sodium hydroxide.

$$\text{Exothermic: } \Delta H < 0 \quad \text{Endothermic: } \Delta H > 0$$

We have defined the reactants as part of the system. The solvent (water in this case) and the calorimeter itself are the surroundings. By measuring the heat absorbed by the solvent and the calorimeter, we can infer the heat change for the reaction. This assumes, of course, that there is no heat loss. Using an apparatus called a calorimeter, the heat lost or absorbed by a reaction in the calorimeter can be calculated using the following equation:

$$\Delta H_{\text{reaction}} = \Delta H_{\text{calorimeter}} + \Delta H_{\text{solution}}$$

For our purposes, we will assume that the heat gained or lost by the calorimeter itself is negligible. Thus, we will simply use the heat lost or gained by the solution.

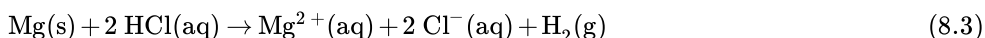
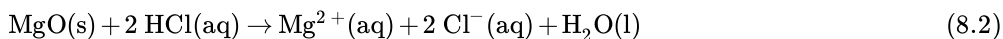
The enthalpy change, ΔH , is determined by measuring the temperature change, ΔT , and the mass of the system. The product of the mass, the specific heat and temperature change of the system will give the enthalpy change for the system. The heat capacity of 1 g of a substance is called its specific heat capacity (4.18 J/g°C for water).

$$\Delta H = (\text{specific heat of water}) \times (\text{grams of solution}) \times (\Delta T)$$

In this experiment we will measure the heat of neutralization of the following reaction:



In addition we will measure the heats of reactions 2 and 3, shown below. Knowing the heat of reaction for 4, we will use Hess's Law and reaction 2-4 to calculate the heat of reaction for 5.



[Pre Lab Video](#)



Safety and Waste Disposal

- All solutions can be disposed of down the drain.

Procedure

Heat of Neutralization: $\text{H}^+(\text{aq}) + \text{OH}^-(\text{aq}) \rightarrow \text{H}_2\text{O}(\text{l})$

Step 1

Obtain a calorimeter consisting of a double Styrofoam cup, a magnetic stirrer and magnet, and the LabQuest with the temperature sensor.

Step 2



Pour 25.0 mL of **2M** sodium hydroxide (NaOH), measured with your graduated cylinder, into your calorimeter.

Step 3



Measure 25.0 mL of **2M** hydrochloric acid (HCl) in clean, graduated cylinder. Clean and dry the temperature sensor and support it in the NaOH solution.



Step 4

Begin *gently* stirring the solution with the magnetic stirrer.



Step 5

Tap the “Start” button. Allow several data points to be collected for 15 seconds to set the initial temperature.

Step 6



With the run ongoing, add the acid (HCl) solution to the base (NaOH),

Step 7

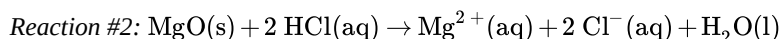
When the temperature starts to decline, tap the “stop” button. Click the “analyze” menu and then selecting “Statistics”, “Temperature” to record the maximum temperature reached and the initial temperature.

Step 8

Empty your calorimeter (the materials can go down the drain), rinse, and dry all parts completely.

Step 9

Repeat the procedure *at least* once more. If your data looks substantially different between the two runs, repeat a third time. Accurate results will be rewarded.



Step 1



Measure out 50.0 mL of **1M** HCl into the calorimeter. **CAUTION:** This is a different HCl concentration than the one used in Reaction #1. Handle the HCl solution with care. It can cause painful burns if it comes in contact with the skin.

Step 2

Weigh out about 0.50 g of magnesium oxide, MgO, on a piece of weighing paper. Record the exact mass used in your data table. **CAUTION:** Avoid inhaling magnesium oxide dust.



Step 3



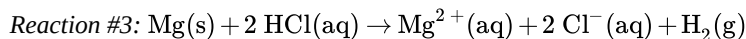
Tap the start button and record the temperature for 15 seconds. Then add the white magnesium oxide powder to the solution. Use the magnetic stirrer to gently stir the cup contents until a maximum temperature has been reached. Record the minimum and maximum temperatures. The waste can go down the drain.

Step 4

Repeat the procedure once more.

Step 5

Clean and dry the calorimeter cup, the stir bar, and the thermometer.



Step 1



Repeat the above experiment using ~0.2 g of Mg, ribbon rather than the MgO powder. The magnesium ribbon has been pre-cut to the proper length. Be sure to record the measured mass of the magnesium and the initial and final temperatures. **CAUTION:** *Do not breathe the vapors produced by this reaction!* The waste can go down the drain.



Processing the Data

All calculations should be *neatly* presented in your laboratory notebook.

Step 1

For each reaction the heat gained or lost by the solution is found using: $q_{\text{soln}} = C_p \bullet m \bullet \Delta T$. [$C_p = 4.18 \frac{\text{J}}{\text{g}^\circ\text{C}}$ and we assume that the solution has the same density as water, $1 \frac{\text{g}}{\text{mL}}$]

Step 2

Calculate the heat gained or lost by the reaction, $q_{\text{rxn}} = -q_{\text{soln}}$

Step 3

Calculate ΔH_{rxn} for each reaction. $\Delta H_{rxn} = \frac{q_{rxn}}{moles}$.

Step 4

Using Hess's law and your experimental ΔH_{rxn} values for reactions 2 and 3, and the calculated value from the table in the back of the book for reaction 4 ($\text{H}_2(\text{g}) + \frac{1}{2} \text{O}_2(\text{g}) \rightarrow \text{H}_2\text{O}(\text{l})$), to calculate the ΔH_{rxn} for reaction 5.

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CHAPTER OVERVIEW

9: Gas Laws

Learning Objectives

- To determine what happens to a gas when it is compressed.
- To determine what happens to a gas when it is heated.
- To determine if different gases behave differently.
- To determine the molar volume of a gas.

In this experiment, we will examine some of the properties of variety of gases, and draw some conclusions about how gases behave based on the pressure, temperature, volume and identity of the gas.

In Part I, we will examine the relationship between the volume and pressure for a gas. In Part II, we will examine the temperature-pressure relationship for air. A second gas will be examined if there is sufficient time to complete the experiment. In Part III, we will determine the molar volume of two different gases and compare those values to the theoretical molar volume of an ideal gas.

Pre Lab Video



Safety and Waste Disposal

The solution can go down the drain with plenty of water.

Experiment

Part I: Pressure-Volume Relationships

Step 1

Prepare the Gas Pressure Sensor and an air sample for data collection. Connect the Gas Pressure Sensor to LabQuest and choose New from the File menu.



Step 2



With the 20 mL syringe disconnected from the Gas Pressure Sensor, move the piston of the syringe until the front edge of the inside black ring is positioned at the 10.0 mL mark.

Step 3



Attach the 20 mL syringe to the valve of the Gas Pressure Sensor.

Step 4



Set up the data-collection mode. On the Meter screen, tap Mode. Change the mode to Events with Entry.

Step 5

Enter the Name (Volume) and Units (mL). Select OK.



Step 6

To obtain the best data possible, you will need to correct the volume readings from the syringe. Look at the syringe; its scale reports its own internal volume. However, that volume is not the total volume of trapped air in your system since there is a little bit of space inside the pressure sensor. To account for the extra volume in the system, you will need to add 0.8 mL to your syringe readings. For example, with a 5.0 mL syringe volume, the total volume would be 5.8 mL. *It is this total volume that you will need for the analysis.*

Step 7



You are now ready to collect pressure and volume data. It is easiest if one person takes care of the gas syringe and another enters volumes. Start data collection. (The green arrow in the bottom left corner or the arrow button on the top of the keyboard).

Step 8



Move the piston so the front edge of the inside black ring is positioned at the 5.0 mL line on the syringe (see Figure 2). Hold the piston firmly in this position until the pressure value displayed on the screen stabilizes.

Step 9



Tap **Keep** and enter **5.8**, the gas volume (in mL) on the screen. Remember, you are adding 0.8 mL to the volume of the syringe for the total volume. Select OK to store this pressure-volume data pair.



Step 10



Continue this procedure using syringe volumes of 10.0, 12.5, 15.0, 17.5, and 20.0 mL adding 0.8 mL each time. Do not remove the syringe in between each volume change.

Step 11



Stop data collection. (The red square on the bottom left of the screen, or the arrow button on the keypad)

Step 12

When data collection is complete, a graph of pressure vs. volume will be displayed. To examine the data pairs on the displayed graph, tap any data point. As you tap each data point, the pressure and volume values are displayed to the right of the graph. Record the pressure and volume data values in your notebook.

Step 13

Repeat steps 1-12 with another gas provided by your instructor if there is time.

PART II – Temperature and Pressure Relationships

Step 1



Prepare a hot water bath. Put about 300 mL of water into a large beaker and place it on a hot plate. Turn the hot plate to a medium setting. Do not use this bath unless the temperature is **below 80 °C.**

Step 2

Prepare an ice-water bath. Place ice in an ice bath tray and add water.



Step 3



Put about 300 mL of room temperature water into another beaker (cold tap).

Step 4



Put about 300 mL of warm tap water into a fourth beaker (35 - 40°C) (hot tap).

Step 5



Prepare the Temperature Probe and Gas Pressure Sensor for data collection. Connect the Gas Pressure Sensor to Channel 1 of LabQuest and the Temperature Probe to Channel 2. Choose New from the File menu.

Step 6



Obtain a rubber-stopper assembly with a piece of heavy-wall plastic tubing connected to one of its two valves. Attach the connector at the free end of the plastic tubing to the open stem of the Gas Pressure Sensor with a clockwise turn. Leave its two-way valve on the rubber stopper open until Step 8.

Step 7



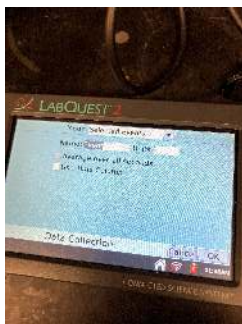
Insert the rubber-stopper assembly into a 125 mL Erlenmeyer flask. **Important:** Twist the stopper into the neck of the flask to ensure a tight fit.

Step 8



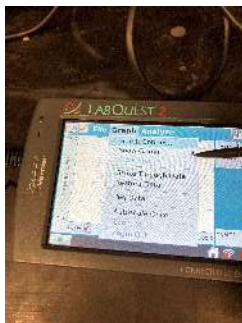
Close the 2-way valve above the rubber stopper—do this by turning the valve handle so it is perpendicular with the valve stem itself. The air sample to be studied is now confined in the flask.

Step 9



On the Meter screen, tap Mode. Change the mode to Selected Events. Select OK.

Step 10



Change the graph settings to display a graph of pressure vs. temperature. Tap the Graph tab. Choose Show Graph ► Graph 1 from the Graph menu. Tap on the x-axis label and choose Temperature.

Step 11



Start data collection. Pressure readings (in kPa) and temperature readings (in °C) are displayed on the screen.

Step 12

Collect pressure vs. temperature data for your gas sample. Place the flask into the ice water bath. Make sure the entire flask is covered. Place the Temperature Probe into the ice water bath. When the temperature and pressure readings have both stabilized, tap



Keep to store the temperature and pressure readings.



Step 13



Repeat Step 12 using the room temperature bath.

Step 14



Repeat Step 12 using the warm water bath.

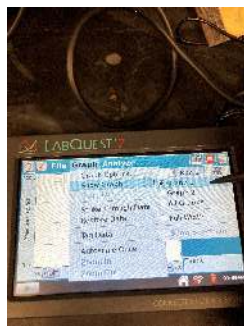
Step 15



CAUTION: Do not burn yourself or the probe wires with the hot plate. Keep all wiring and tubing away from the hot plate surface. **It will melt the plastic.** Place the temperature probe in the hot water bath. After the Temperature Probe has been in the hot water for a few seconds, place the flask into the hot water bath and repeat Step 12. Stop data collection and remove the flask and the Temperature Probe.

Step 16

Examine the data points along the displayed graph of pressure vs. temperature ($^{\circ}\text{C}$). To examine the data pairs on the displayed graph, tap any data point. As you tap each data point, the temperature and pressure values are displayed to the right of the graph. Record the data pairs in your data table.



Step 17

Repeat these experiments with the second gas if your instructor directs you to do this.

Part III – Molar Volume of Gas

Step 1



Attach the pressure probe to Channel 1 on the LabQuest, and attach the temperature probe to Channel 2. On the right hand screen, change the mode to “Time Based.” Set the duration for 10 minutes and the rate at 3 samples per minute.

Step 2

Determine the *exact* volume of gas held in your 125 mL Erlenmeyer flask (after accounting for the stopper and the acid you will be adding).

Step 3

Prepare a room temperature water bath in a 600 mL beaker. Make sure that there is enough water to cover the entire flask, but be careful that displaced water does not spill out.

Step 4



Weigh about 0.01 g of magnesium and record the exact mass in your lab notebook. Put the magnesium in your 125 mL flask.

Step 5

Measure out about 6 mL of 1 M HCl into a small beaker, and draw 5 mL into the syringe from part 1. Attach the syringe to the two way stopper. Make sure that the valve connected to the syringe is closed (turn the knob perpendicular).



Step 6



Put the stopper in the flask, and place the flask in the water bath. Also insert the temperature probe into the water bath.

Step 7



Press the green play button on the Vernier. 20 seconds after pressing “play,” open the valve below the syringe and empty the HCl into the flask. Return the syringe to the 5 mL position and close the valve.



Step 8

Allow data collection to continue for the next 10 minutes.

Step 9

Analyze the data. From the “Analyze” menu, select “statistics” and then “pressure.” Repeat for temperature. Record the minimum pressure and maximum pressure. If the pressure peaked and then dropped again before 1 minute had passed, click the table icon and determine the maximum pressure from the table. **Do not use the pressure “spike” as your maximum pressure.** Record the mean temperature.

PART IV – Calculations

Step 1

Using the data gathered from Part I and Part II, create graphs using excel. Be sure to convert your temperature in Celsius to Kelvin before creating the graph for part II.

Step 2

For Part III, write out the balanced chemical equation for the reaction of magnesium with hydrochloric acid.

Step 3

Calculate the change in pressure.

Step 4

Use the equation for reaction and the amounts of reactants used to determine the limiting reactants and the molar amount of gas produced.

Step 5

Using STP (Standard temperature and pressure) values and the conditions in your trial, determine the volume of the gas you produced at STP using the equation $\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}$

Step 6

Divide your calculated STP volume by the number of moles of gas produced to determine the molar volume of gas at STP. Compare to the theoretical molar volume of an ideal gas ($22.4 \frac{L}{mol}$).

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CHAPTER OVERVIEW

10: Intermolecular Forces

Learning Objectives

- To research data in the chemical literature and evaluate the sources.
- To examine the relationship between molecular structure and evaporation.
- To examine the relationship between molecular structure and boiling point.
- To examine the relationship between evaporation and boiling point.
- To graph data and use these plots to interpret chemical data and the relationship to the properties of the compounds studied.
- To orally present the results of the experiment to an audience.

Pre Lab Video



Pre-Lab Assignment

Before coming to lab, you need to look up the boiling points of the following compounds: methanol (CH_3OH), ethanol ($\text{CH}_3\text{CH}_2\text{OH}$), 1-propanol ($\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$), 1-pentanol ($\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{OH}$), pentane (C_5H_{12}), hexane (C_6H_{14}), heptane (C_7H_{16}), octane (C_8H_{18}), acetone (a.k.a. propanone) (CH_3COCH_3), 2-butanone (a.k.a. methyl ethyl ketone) ($\text{CH}_3\text{COCH}_2\text{CH}_3$), 2-pentanone ($\text{CH}_3\text{COCH}_2\text{CH}_2\text{CH}_3$), water (a.k.a. dihydrogen oxide) (H_2O). You will need find the data from two different sources and cite the sources from which you obtained the information. Note: it is OK if the sources do not agree; we will discuss the data collection in pre-lab.

You should also draw the Lewis structures of these molecules. For hexane, pentane, heptane and octane, draw the linear structure (normally referred as *n-alkane*) where each carbon atom is attached to two other carbon atoms. You should also have calculated the molecular weights for all of the compounds as well.

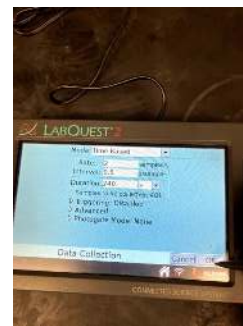
Experiment Week 1

Caution

The compounds used in this experiment are flammable and poisonous. Avoid inhaling their vapors. Be sure there are no open flames in the lab during this experiment.

Step 1

Set-up the LabQuest. Connect the Temperature Probes to LabQuest and choose New from the File menu. On the Meter screen, tap Duration. Change the data-collection length to 240 seconds. Select OK. Data collection will last 4 minutes.

Two yellow-tipped probes or sensors are shown lying on a dark, textured surface. The probes have black handles and yellow tips.

<https://chem.libretexts.org/@go/page/316205>

Step 6

Monitor the temperature as it is being recorded and let the experiment run for the full 240 seconds.

Step 7



Once the run has stopped, you will need to obtain the minimum and maximum temperatures. To do this, click on the “analyze” menu and choose statistics. Select one of your runs. The statistics (including minimum and maximum temperatures) will appear on the right. Record these in your notebook with the identity of your liquid. Repeat for your other run. You may need to scroll the window on the right down to see both data sets.

Step 8

Move to the next station when it is available and repeat steps 3 thru 7.

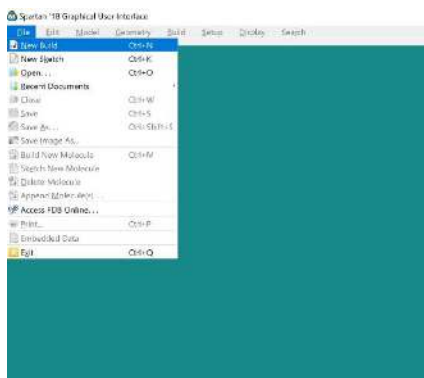
Step 9

Repeat this procedure for each of the 12 liquids. Do not contaminate the liquids.

Data Analysis:

Using Spartan, you will draw the molecules and calculate their dipole moments. Collect this data in your lab notebook.

Step 1



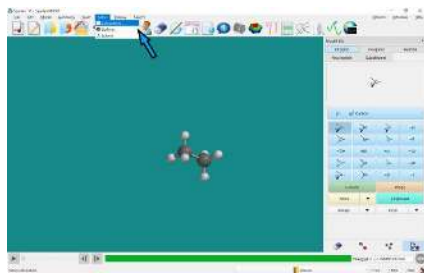
Build the molecule in Spartan as you did previously.

Step 2



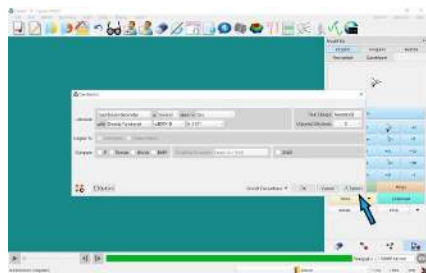
Minimize the energy as you did before with the button (down arrow/E)

Step 3



In the “Setup” menu, choose “Calculation”

Step 4

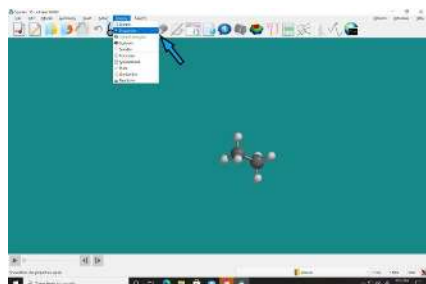


Leave all of the selections as they are and click on “submit”. You must click submit. Do not click OK.

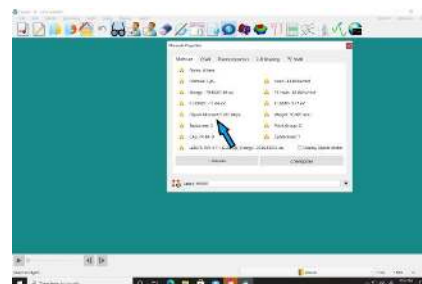
Step 5

Choose a file name and let it run. Click OK (should need to do so twice).

Step 6



After it says it is done, go to “display” menu and select “Properties”. The dipole moment should be displayed in the box. Record this in your notebook.



Step 7

Repeat for the other molecules.

Post Lab

You will prepare four graphs containing the following plots:

- ΔT versus boiling point for all of the compounds together on one plot line.
- Boiling point versus dipole moment with a plot line for each class of compound (alkanes, alcohols and ketones) This requires that there be three different plot lines on one graph with a trend line for each line.
- Boiling point versus MW with a plot line for each class of compound (alkanes, alcohols and ketones) - three different plot lines on one graph.
- ΔT versus MW for each class of compound (alkanes, alcohols and ketones) – three lines on one graph.

Experiment Week 2:

You will be given a set of compounds that you have not yet measured. Measure their ΔT as you did in Week 1. Follow the same procedure, recording your observations and the value of your ΔT . You will need to then look up the boiling points from the same site you obtained the previous set (if they were reasonable), draw the Lewis structures and calculate the dipole moments. You will then spend the remainder of the lab period working with your lab partner on the following:

1. Examine how your new compounds fit into your previous data set. Do they make sense? Explain why or why not.
2. Prepare a brief presentation (as a pair) to be given during the laboratory session next week that explains your approach to answering the above question. You should use the data from the previous week to analyze your data and what you know about the structure of the molecule and intermolecular forces to explain your data. See the report guidelines for a rubric that will be used to grade the presentation.

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CHAPTER OVERVIEW

11: Preparatory Sheets

[11.1: Density and Graphing Prep](#)

[11.2: Halogens Prep Sheet](#)

[11.3: Lewis Structure Prep Sheet](#)

[11.4: Spectroscopy Prep Sheet](#)

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11.1: Density and Graphing Prep

Materials

- 10 balances

Chemicals

- Saturated sodium chloride solution (can be made with table salt)

SDS Info

- Sodium Chloride – None

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11.2: Halogens Prep Sheet

Materials

- Box of new beral pipets
- 6 inch test tubes (100)
- Parafilm

Chemicals

All of the solutions are aqueous solutions unless otherwise specified and should have test tube and pipette attached.

5 labeled jars/bottles

- 10 g NaCl
- 10 g NaBr
- 10 g NaI
- 100 mL of cyclohexane

In the hood: 2 bottles of each

- 200 mL of 0.1M NaCl
- 200 mL of 0.1 M NaBr
- 200 mL of 0.1 M NaI
- 500 mL of chlorine water (25 ml of bleach with 10 mL of 6M HCl) dilute to 500 mL. Allow to sit uncovered (in hood) until it stops bubbling. Store in tightly closed amber container with parafilm around the top. Shelf life of about 1 week. Store in 2 separate bottles – **Label corrosive - Store in refrigerator**
- 400 mL of bromine water (2 mL of Br₂ in 400 mL of water) – Br₂ is highly corrosive to skin! Stir on hot plate overnight covered with parafilm. Store in 2 separate amber bottles with parafilm around the top – **Label corrosive - Store in refrigerator**
- 500 mL of iodine water (0.4 g of I₂ in 80 mL of MeOH and add 420 mL of water) Store in 2 separate amber bottles with parafilm around the top – **Label corrosive - Store in refrigerator**

2 waste bottles

SDS Info

- Sodium Chloride – None
- Sodium Bromide – None
- Sodium Iodide – Exclamation point – skin or eye irritant; Environmental – aquatic toxicity
- Chlorine – Exclamation point – skin or eye irritant; Exclamation point – respiratory tract irritant; Gas cylinder – gases under pressure, Flame over circle – Oxidizer; Environmental – aquatic toxicity; Skull and crossbones – acute toxicity
- Bromine – Corrosion – Skin corrosion/burns; Corrosion – eye damage; Environmental – aquatic toxicity; Skull and crossbones – acute toxicity
- Iodine – Health Hazard – Target Organ Toxicity; Exclamation point – skin or eye irritant; Exclamation point – respiratory tract irritant; Environmental – aquatic toxicity
- Cyclohexane – Health Hazard – may be fatal if swallowed or enters airways, Flame – flammable; Exclamation Point – skin or eye irritant; Environmental – aquatic toxicity

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11.3: Lewis Structure Prep Sheet

No prep for this week.

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11.4: Spectroscopy Prep Sheet

Materials

- 10 High voltage lamp holders – 2 per bench
 - Into each lamp holder place one H₂, and one other lamp either Kr, Ar, He, Xe, or Ne.
 - Spectrophotometer with all of the appropriate cables – power, computer and fiber optic.
 - 10 handheld spectrosopes, 2 per bench
-

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